DECISION SUPPORT SYSTEM (DSS)

ELECTRE IV

TEACHING TEAM

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DEFINITION

- Elimination Et Choix Traduisant la Réalité
- ELECTRE-Elimination and Choice Expressing Reality
- It is the concept of **autorank** to determine the pairing comparison of existing alternatives
- The principle of ELECTRE calculation is that the dominance of one alternative over another
- ELECTRE is used for cases where there are many alternatives but few criteria

DEVELOPMENT OF THE ELECTRE METHOD

- This method was developed in the mid-1960s in Europe by Bernard Roy.
- The ELECTRE method is used to select the best Actions from an existing set of Actions by applying, choosing, ranking, and sorting.
- This method developed through a number of versions consisting of ELECTRE I, ELECTRE II, ELECTRE III, and ELECTRE IV.
- The ELECTRE I method is designed for selection while the ELECTRE II is used for ranking. Both versions are simple types of criteria while the other version uses pseudo-criteria.
- ELECTRE IV is a version based on pseudo-criteria that aims to rank alternatives, with or without introducing weighting of any criteria.

DEVELOPMENT OF THE ELECTRE METHOD

- ELECTRE I

This method is built for multicriteria choice problems: The goal is to be able to get a subset of N of those actions like any action that is not on the N outrank with at least one N action.

The threshold value is the same for all criteria

- ELECTRE II

The main difference between this method and ELECTRE I lies in defining two outranking relationships instead of one: strong outranking and weak outranking.

The ELECTRE II method, aims to make a ranking from best to worst (problem ranking).

The ranking procedure in the ELECTRE II method is preceded by the formation of a directional graph as a representation of the outranking relationship, then ranking is carried out based on the graph by looking for the average value of both the concordance and discordance index rankings.

The threshold value is the same for all criteria

- ELECTRE III

This version of the method uses pseudo criteria whose threshold values are not the same for all criteria.

- ELECTRE IV

This version of the method uses pseudo criteria whose threshold values are not the same for all criteria.

This version of the method aims to rank actions, but without introducing any weighting criteria.

So that in this version it can be determined alternative rankings, with or without introducing weighting of any criteria

DEVELOPMENT OF THE ELECTRE METHOD

- Conclusion:
- The ELECTRE I method is designed for selection while the ELECTRE II is used for ranking. Both versions use a simple type of criterion, in the sense that the threshold value is the same for all criteria while versions III and IV use pseudo-criteria where the threshold value is not the same for all criteria.

ELECTRE METHOD STEPS

- Normalization of the decision matrix
- Normalized matrix weighting
- Determining concordance and dissordance sets
- Calculating concordance and disordance matrices
- Determine the dominant matrix of concordance and dissordance
- Define the aggregate dominance matrix
- Elimination of less favourable alternatives

1. NORMALIZATION OF THE DECISION MATRIX

- Attributes are converted to comparable values.
- Normalization can be done with the formula:
  - BENEFIT → $r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x^2{}_{ij}}}$ for *i = 1, 2, 3, ... , m* and *j = 1, 2, 3, ... ,*
  - COST→ $r_{ij} = 1 - \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x^2{}_{ij}}}$

So that the normalization result matrix is obtained:

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & & & \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix}$$

Example of normalization of the decision matrix

$$X = \begin{bmatrix} 4 & 4 & 5 & 3 & 3 \\ 3 & 3 & 4 & 2 & 3 \\ 5 & 4 & 2 & 2 & 2 \end{bmatrix}$$

$$r_{11} = \frac{x_{11}}{\sqrt{\sum_{i=1}^{m} x^2{}_{i1}}} = \frac{4}{\sqrt{4^2 + 3^2 + 5^2}} = \frac{4}{7{,}071} = 0{,}5657$$

$$r_{12} = \frac{x_{12}}{\sqrt{\sum_{i=1}^{m} x^2{}_{i2}}} = \frac{4}{\sqrt{4^2 + 3^2 + 4^2}} = \frac{4}{6{,}403} = 0{,}6247$$

$$R = \begin{bmatrix} 0{,}5657 & 0{,}6247 & 0{,}7454 & 0{,}7276 & 0{,}6396 \\ 0{,}4243 & 0{,}4685 & 0{,}5963 & 0{,}4851 & 0{,}6396 \\ 0{,}7071 & 0{,}6247 & 0{,}2981 & 0{,}4851 & 0{,}4264 \end{bmatrix}$$

2. WEIGHTING ON THE NORMALIZED MATRIX

- Once normalized, each column of the matrix R is multiplied by the weight of W so that

$$V = R \, . \, W$$

$$\begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & & & \\ v_{m1} & v_{m2} & \cdots & v_{mn} \end{bmatrix} = \begin{bmatrix} w_1 r_{11} & w_2 r_{12} & \cdots & w_n r_{1n} \\ w_1 r_{21} & w_2 r_{22} & \cdots & w_n r_{2n} \\ \vdots & & & \\ w_1 r_{m1} & w_2 r_{m2} & \cdots & w_n r_{mn} \end{bmatrix}$$

$$W = \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & & & \\ 0 & 0 & \cdots & w_n \end{bmatrix}$$

Example of normalized matrix weighting

- $V = R\,.W$
- $= \begin{bmatrix} 0{,}5657 & 0{,}6247 & 0{,}7454 & 0{,}7276 & 0{,}6396 \\ 0{,}4243 & 0{,}4685 & 0{,}5963 & 0{,}4851 & 0{,}6396 \\ 0{,}7071 & 0{,}6247 & 0{,}2981 & 0{,}4851 & 0{,}4264 \end{bmatrix} . \begin{bmatrix} 5 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$
- $= \begin{bmatrix} 2{,}8285 & 1{,}8741 & 2{,}9816 & 2{,}9104 & 1{,}2792 \\ 2{,}1215 & 1{,}4055 & 2{,}3852 & 1{,}9404 & 1{,}2792 \\ 3{,}5366 & 1{,}8741 & 1{,}1924 & 1{,}9404 & 0.8528 \end{bmatrix}$

3. DETERMINING CONCORDANCE AND DISSORDANCE SETS

- For each pair of values k and l (k, l = 1, 2, 3, ... , m and k≠1)
- The J set of criteria is divided into 2 subsets: concordance and discordance
- Concordance

$$\boxed{C_{kl} = \{j, v_{kj} \geq v_{ij}\}, for\ j = 1, 2, 3, \dots, n}$$

- Disordance

$$\boxed{D_{kl} = \{j, v_{kj} < v_{ij}\}, for\ j = 1, 2, 3, \dots, n}$$

Example of concordance and dissordance set calculation

$$\begin{bmatrix} 2{,}8285 & 1{,}8741 & 2{,}9816 & 2{,}9104 & 1{,}2792 \\ 2{,}1215 & 1{,}4055 & 2{,}3852 & 1{,}9404 & 1{,}2792 \\ 3{,}5366 & 1{,}8741 & 1{,}1924 & 1{,}9404 & 0.8528 \end{bmatrix}$$

$$C_{kl} = \{j, v_{kj} \geq v_{ij}\}, for\ j = 1, 2, 3, \dots, n$$

$C_{12} = \{j, v_{1j} \geq v_{2j}\}, for\ j = 1, 2, 3, \dots, 5$
={1,2,3,4,5}

$C_{13} = \{j, v_{1j} \geq v_{3j}\}, for\ j = 1, 2, 3, \dots, 5$
={2,3,4,5}

$C_{21} = \{j, v_{1j} \geq v_{1j}\}, for\ j = 1, 2, 3, \dots, 5$
={5}

$C_{23} = \{j, v_{2j} \geq v_{3j}\}, for\ j = 1, 2, 3, \dots, 5$
={3,4,5}

$C_{31} = \{j, v_{3j} \geq v_{1j}\}, for\ j = 1, 2, 3, \dots, 5$
={1,2}

$C_{32} = \{j, v_{3j} \geq v_{2j}\}, for\ j = 1, 2, 3, \dots, 5$
={1,2,4}

$$D_{kl} = \{j, v_{kj} < v_{ij}\}, for\ j = 1, 2, 3, \dots, n$$

$D_{12} = \{j, v_{1j} < v_{2j}\}, for\ j = 1, 2, 3, \dots, 5$
={ }

$D_{13} = \{j, v_{1j} < v_{3j}\}, for\ j = 1, 2, 3, \dots, 5$
={1}

$D_{21} = \{j, v_{1j} < v_{1j}\}, for\ j = 1, 2, 3, \dots, 5$
={1,2,3,4}

$D_{23} = \{j, v_{2j} < v_{3j}\}, for\ j = 1, 2, 3, \dots, 5$
={1,2}

$D_{31} = \{j, v_{3j} < v_{1j}\}, for\ j = 1, 2, 3, \dots, 5$
={3,4,5}

$D_{32} = \{j, v_{3j} < v_{2j}\}, for\ j = 1, 2, 3, \dots, 5$
={3,5}

4. A. CALCULATING THE CONCORDANCE MATRIX

- Calculated by showing the number of weights included in the concordance set

$$c_{kl} = \sum_{j \in c_{kl}} w_j$$

Example of concordane matrix calculation

$c_{12} = w_1 + w_2 + w_3 + w_4 + w_5$
= 5 + 3 + 4 + 4 + 2 = 18

$c_{13} = w_2 + w_3 + w_4 + w_5$
= 3 + 4 + 4 + 2 = 13

$c_{21} = w_5$
= 2

$c_{23} = w_3 + w_4 + w_5$
= 4 + 4 + 2 = 10

$c_{31} = w_1 + w_2$
= 5 + 3 = 8

$c_{32} = w_1 + w_2 + w_4$
= 5 + 3 + 4 = 12

$$\begin{bmatrix} - & 18 & 13 \\ 2 & - & 10 \\ 8 & 12 & - \end{bmatrix}$$

4. B. CALCULATING THE DISORDANCE MATRIX

- It is done by dividing the maximum difference of the criteria included in the discordance subset by the maximum difference in the value of all existing criteria

$$d_{kl} = \frac{max\{|v_{kj} - v_{lj}|\}_{j \in D_{kl}}}{max\{|v_{kj} - v_{lj}|\}_{\forall_j}}$$

Example of disordance matrix calculation

$$\begin{bmatrix} 2{,}8285 & 1{,}8741 & 2{,}9816 & 2{,}9104 & 1{,}2792 \\ 2{,}1215 & 1{,}4055 & 2{,}3852 & 1{,}9404 & 1{,}2792 \\ 3{,}5366 & 1{,}8741 & 1{,}1924 & 1{,}9404 & 0.8528 \end{bmatrix}$$

$$\begin{bmatrix} - & 0 & 0{,}3951 \\ 1 & - & 1 \\ 1 & 12 & - \end{bmatrix}$$

$$d_{12} = \frac{max\{|v_{1j} - v_{2j}\}_{j \in D_{12}}}{max\{|v_{1j} - v_{2j}\}_{\forall j}}$$

$$= \frac{max\{0\}}{max\{|2{,}8285-2{,}1215|;|1{,}8741-1{,}2055|;|2{,}9816-2{,}3852|;|2{,}9104-1{,}9404|;|1{,}2792-1{,}2792|\}}$$

$$= \frac{max\{0\}}{max\{0{,}7070;0{,}4686;0{,}5964;0{,}9700;0\}} = 0$$

$$d_{13} = \frac{max\{|v_{1j} - v_{3j}\}_{j \in D_{13}}}{max\{|v_{1j} - v_{3j}\}_{\forall j}}$$

$$= \frac{max\{2{,}8285-3{,}5366\}}{max\{|2{,}8285-3{,}5355|;|1{,}8741-1{,}8741|;|2{,}9816-1{,}1924|;|2{,}9104-1{,}9404|;|1{,}2792-0{,}8528|\}}$$

$$= \frac{max\{0{,}7081\}}{max\{0{,}7070;0;1{,}7892;0{,}9701;0{,}4264\}} = 0{,}3957$$

$$D_{kl} = \{j, v_{kj} < v_{ij}\}, for\ j = 1, 2, 3, \dots, n$$

$D_{12} = \{j, v_{1j} < v_{2j}\}, for\ j = 1, 2, 3, \dots, 5$
={ }

$D_{13} = \{j, v_{1j} < v_{3j}\}, for\ j = 1, 2, 3, \dots, 5$
={1}

$D_{21} = \{j, v_{1j} < v_{1j}\}, for\ j = 1, 2, 3, \dots, 5$
={1,2,3,4}

$D_{23} = \{j, v_{2j} < v_{3j}\}, for\ j = 1, 2, 3, \dots, 5$
={1,2}

$D_{31} = \{j, v_{3j} < v_{1j}\}, for\ j = 1, 2, 3, \dots, 5$
={3,4,5}

$D_{32} = \{j, v_{3j} < v_{2j}\}, for\ j = 1, 2, 3, \dots, 5$
={3,5}

5. A. CALCULATING THE CONCORDANCE DOMINANT MATRIX

- The concordance dominant matrix is constructed using thresholds
- This is done by comparing the value of each concordance matrix element with the threshold value
- $c_{kl} \geq \underline{c}$
- With a threshold value($\underline{c}$)
- $\underline{c} = \frac{\sum_{k=1}^{m} \sum_{l=1}^{m} c_{kl}}{m(m-1)}$
- Thus, the value of the matrix F can be calculated
- $f_{kl} = \begin{cases} 1, jika\ c_{kl} \geq \underline{c} \\ 0, jika\ c_{kl} < \underline{c} \end{cases}$

Example of concordance dominant matrix calculation

$$\underline{c} = \frac{\sum_{k=1}^{m}\sum_{l=1}^{m} c_{kl}}{m(m-1)}$$

$$= \frac{18 + 13 + 2 + 10 + 8 + 12}{3(3-1)} = \frac{63}{6} = 10{,}5$$

$$\begin{bmatrix} - & 18 & 13 \\ 2 & - & 10 \\ 8 & 12 & - \end{bmatrix}$$

The elements of the matrix F are defined as follows:

$$f_{kl} = \begin{cases} 1\,, jika\ c_{kl} \geq \underline{c} \\ 0, jika\ c_{kl} < \underline{c} \end{cases}$$

So that the dominant matrix concordance: $F = \begin{bmatrix} - & 1 & 1 \\ 0 & - & 0 \\ 0 & 1 & - \end{bmatrix}$

5. B. Calculating the dominant matrix of DISORDANCE

- Matrix G is the dominant matrix of discordence

$$\underline{d} = \frac{\sum_{k=1}^{m} \sum_{l=1}^{m} d_{kl}}{m(m-1)}$$

- With matrix elements defined by:

$$g_{kl} = \begin{cases} 1\,, jika\ d_{kl} \geq \underline{d} \\ 0, jika\ d_{kl} < \underline{d} \end{cases}$$

Example of discordance dominant matrix calculation

$$\underline{d} = \frac{\sum_{k=1}^{m}\sum_{l=1}^{m} d_{kl}}{m(m-1)}$$

$$= \frac{0 + 0{,}3951 + 1 + 1 + 1 + 1}{3(3-1)} = \frac{4{,}3951}{6} = 0{,}7325$$

$$\begin{bmatrix} - & 0 & 0{,}3951 \\ 1 & - & 1 \\ 1 & 12 & - \end{bmatrix}$$

The elements of the matrix F are defined as follows:

$$g_{kl} = \begin{cases} 1\ , jika\ d_{kl} \geq \underline{d} \\ 0, jika\ d_{kl} < \underline{d} \end{cases}$$

So that the dominant matrix is dissordanced: $G = \begin{bmatrix} - & 0 & 0 \\ 1 & - & 1 \\ 1 & 1 & - \end{bmatrix}$

6. Determining the dominant aggregate of the matrix

- Matrix E is obtained from the multiplication between matrix element F and matrix element G

$$e_{kl} = f_{kl} \ x \ g_{kl}$$

Example of matrix dominant aggregate calculation

$$e_{kl=} f_{kl}\ x\ g_{kl}$$

$$e_{12} = f_{12}\ x\ g_{12} = 1\ x\ 0 = 0$$

$$e_{13} = f_{13}\ x\ g_{13} = 1\ x\ 0 = 0$$

$$e_{21} = f_{21}\ x\ g_{21} = 0\ x\ 1 = 0$$

$$e_{23} = f_{23}\ x\ g_{23} = 0\ x\ 1 = 0$$

$$e_{31} = f_{31}\ x\ g_{31} = 1\ x\ 0 = 0$$

$$e_{32} = f_{32}\ x\ g_{32} = 1\ x\ 1 = 1$$

Aggregate Matriks Dominan

$$\mathrm{E} = \begin{bmatrix} - & 0 & 0 \\ 0 & - & 0 \\ 0 & 1 & - \end{bmatrix}$$

$$F = \begin{bmatrix} - & 1 & 1 \\ 0 & - & 0 \\ 0 & 1 & - \end{bmatrix} \qquad G = \begin{bmatrix} - & 0 & 0 \\ 1 & - & 1 \\ 1 & 1 & - \end{bmatrix}$$

7. Elimination of less favourable alternatives

- Matrix E is obtained in the order of choice of each alternative, i.e. if $\mathbf{E_{kl} = 1}$ then the alternative $\mathbf{A_k}$ **is a better alternative to** $\mathbf{A_l}$
- So that the row in the matrix E which has a sum of $\mathbf{E_{kl} = 1}$ the least can be eliminated
- Thus, the best alternative is the one that dominates the other alternatives

CASE STUDY EXAMPLE

CASE STUDY

- A company wants to build a building that will be used as a place to temporarily store its production.
- There are 3 locations that will be alternatives, namely:
  - A1: Soekarno Hatta
  - A2: Karangploso
  - A3: Pakis

CASE STUDY

- There are 5 criteria that are used as a reference in decision-making, namely:
- C1: Distance to nearest market (km)
- C2: Density of people around the site (people/km2)
- C3: Distance from factory (km)
- C4: Distance to existing Warehouse (km)
- C5: Land price for location (Rp/m2)

CASE STUDY

- The level of importance of each criterion, is also assessed with 1 to 5, namely:
- 1 = Very low
- 2 = Low
- 3 = Enough
- 4 = Height
- 5 = Very High
- Decision-making gives the following weight to preferences:
- W={5,3,4,4,2}

CASE STUDY

- Evaluate each alternative in each criterion:

Alternatif	Kriteria				
	C1	**C2**	**C3**	**C4**	**C5**
A1	0,75	2000	18	50	500
A2	0,50	1500	20	40	450
A3	0,90	2050	35	35	800

1. Creating a normalized matrix, R

Rumus : $r_{ij} = \dfrac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}$

Alternatif	Kriteria				
	C_1	C_2	C_3	C_4	C_5
A_1	0,75	2000	18	50	500
A_2	0,50	1500	20	40	450
A_3	0,90	2050	35	35	800

$$x_1 = \sqrt{0,75^2 + 0,50^2 + 0,90^2} = 1,273$$

$$r_{11} = \frac{0,75}{1,273} = 0,589$$

$$r_{21} = \frac{0,50}{1,273} = 0,392$$

$$r_{31} = \frac{0,90}{1,273} = 0,706$$

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$$R = \begin{bmatrix} 0,589 & 0,618 & 0,407 & 0,685 & 0,478 \\ 0,392 & 0,463 & 0,453 & 0,548 & 0,430 \\ 0,706 & 0,634 & 0,792 & 0,479 & 0,765 \end{bmatrix}$$

2. Weighting on the normalized matrix

Rumus : $V = R \times W$

$$R = \begin{bmatrix} 0{,}589 & 0{,}618 & 0{,}407 & 0{,}685 & 0{,}478 \\ 0{,}392 & 0{,}463 & 0{,}453 & 0{,}548 & 0{,}430 \\ 0{,}706 & 0{,}634 & 0{,}792 & 0{,}479 & 0{,}765 \end{bmatrix}$$

$$W = 5 \quad 3 \quad 4 \quad 4 \quad 2$$

$$V = \begin{bmatrix} 2{,}945 & 1{,}854 & 1{,}628 & 2{,}740 & 0{,}956 \\ 1{,}960 & 1{,}389 & 1{,}812 & 2{,}192 & 0{,}860 \\ 3{,}530 & 1{,}902 & 3{,}168 & 1{,}916 & 1{,}530 \end{bmatrix}$$

3. Determining the Concordance and Discordance Sets

a. Concordance

A criterion in an alternative includes Concordance if:

$$C_{kl} = \{ j \mid v_{kj} \geq v_{ij} \} \text{ untuk } j = 1,2,3 \ldots n$$

C12 = {1,2,4,5}
C13 = {4}
C21 = {3}
C23 = {4}
C31 = {1,2,3,5}
C32 = {1,2,3,5}

$$V = \begin{bmatrix} 2{,}945 & 1{,}854 & 1{,}628 & 2{,}740 & 0{,}956 \\ 1{,}960 & 1{,}389 & 1{,}812 & 2{,}192 & 0{,}860 \\ 3{,}530 & 1{,}902 & 3{,}168 & 1{,}916 & 1{,}530 \end{bmatrix}$$

3. Determining the Concordance and Discordance Set

b. Discordanse

A criterion in an alternative includes Discordance if:

$$D_{kl} = \{ j \mid v_{kj} < v_{ij} \} \text{ untuk } j = 1,2,3 \ldots n$$

C12 = {3}

C13 = {1,2,3,5}

C21 = {1,2,4,5}

C23 = {1,2,3,5}

C31 = {4}

C32 = {4}

$$V = \begin{bmatrix} 2,945 & 1,854 & 1,628 & 2,740 & 0,956 \\ 1,960 & 1,389 & 1,812 & 2,192 & 0,860 \\ 3,530 & 1,902 & 3,168 & 1,916 & 1,530 \end{bmatrix}$$

4. Determining the Concordance and Discordance Matrix

a. Concordance

$C_{12} = w_1+w_2+w_4+w_5 = 5+3+4+2 = 14$

$C_{13} = w_4 = 4$

$C_{21} = w_3 = 4$

$C_{23} = w_4 = 4$

$C_{31} = w_1+w_2+w_3+w_5 = 5+3+4+2 = 14$

$C_{32} = w_1+w_2+w_3+w_5 = 5+3+4+2 = 14$

$$C = \begin{bmatrix} - & 14 & 4 \\ 4 & - & 4 \\ 14 & 14 & - \end{bmatrix}$$

$W = 5 \quad 3 \quad 4 \quad 4 \quad 2$

C12 = {1,2,4,5}
C13 = {4}
C21 = {3}
C23 = {4}
C31 = {1,2,3,5}
C32 = {1,2,3,5}

4. Determining the Concordance and Discordance Matrix

b. Discordance

$$d_{kl} = \frac{max\{|v_{kj} - v_{lj}|\}_{j \in D_{kl}}}{max\{|v_{kj} - v_{lj}|\}_{\forall j}}$$

$$V = \begin{bmatrix} 2{,}945 & 1{,}854 & 1{,}628 & 2{,}740 & 0{,}956 \\ 1{,}960 & 1{,}389 & 1{,}812 & 2{,}192 & 0{,}860 \\ 3{,}530 & 1{,}902 & 3{,}168 & 1{,}916 & 1{,}530 \end{bmatrix}$$

$$\begin{aligned} d_{12} &= \frac{\max\{|1{,}628 - 1{,}812|\}}{\max\{|2{,}945 - 1{,}960|;|1{,}854 - 1{,}389|;|1{,}628 - 1{,}812|;|2{,}740 - 2{,}192|;|0{,}956 - 0{,}860|\}} \\ &= \frac{\max\{0{,}184\}}{\max\{0{,}985;0{,}465;0{,}184;0{,}548;0{,}096\}} \\ &= \frac{0{,}184}{0{,}985} \\ &= 0{,}186 \end{aligned}$$

$$\begin{aligned} d_{13} &= \frac{\max\{|2{,}945 - 3{,}530|;|1{,}854 - 1{,}902|;|1{,}628 - 3{,}168|;|0{,}956 - 1{,}530|\}}{\max\{|2{,}945 - 3{,}530|;|1{,}854 - 1{,}902|;|1{,}628 - 3{,}168|;|2{,}740 - 1{,}916|;|0{,}956 - 1{,}530|\}} \\ &= \frac{\max\{0{,}585;0{,}048;1{,}540;0{,}574\}}{\max\{0{,}585;0{,}048;1{,}540;0{,}824;0{,}574\}} \\ &= \frac{1{,}540}{1{,}540} \\ &= 1 \end{aligned}$$

C12 = {3}
C13 = {1,2,3,5}
C21 = {1,2,4,5}
C23 = {1,2,3,5}
C31 = {4}
C32 = {4}

Hasil Matrik Discordance

$$D = \begin{bmatrix} - & 0{,}186 & 1 \\ 1 & - & 1 \\ 0{,}535 & 0{,}535 & - \end{bmatrix}$$

5. Calculating the Dominant Matrix, Concordance and Discordance

a. Concordance $C_{kl} \Rightarrow \geq \underline{c}$

$f_{kl} = 1$ jika $c_{kl} \geq \underline{c}$ dan $f_{kl} = 0$ jika $c_{kl} < \underline{c}$

Nilai Threshold ($\underline{c}$) adalah :

$$\underline{c} = \frac{\sum_{k=1}^{n}\sum_{l=1}^{n} c_{kl}}{m(m-1)}$$

$$\underline{c} = \frac{14+4+4+4+14+14}{3(3-1)}$$

$$\underline{c} = \frac{54}{6}$$

$$\boxed{\underline{c} = 9}$$

$$C = \begin{bmatrix} - & 14 & 4 \\ 4 & - & 4 \\ 14 & 14 & - \end{bmatrix}$$

m= jumlah alternatif

Matrik dominan concordance

$$F = \begin{bmatrix} - & 1 & 0 \\ 0 & - & 0 \\ 1 & 1 & - \end{bmatrix}$$

5. Calculating the Dominant Matrix, Concordance and Discordance

b. Discordence $d_{kl} = \geq \underline{d}$

$g_{kl} = 1$ jika $d_{kl} \geq \underline{d}$ dan $g_{kl} = 0$ jika $d_{kl} < \underline{d}$

Nilai Threshold ($\underline{d}$) adalah :

$$\underline{d} = \frac{\sum_{k=1}^{n}\sum_{l=1}^{n} d_{kl}}{m(m-1)}$$

$$\underline{d} = \frac{0{,}186+1+1+1+0{,}535+0{,}535}{3(3-1)}$$

$$\underline{d} = \frac{4{,}256}{6}$$

$$\boxed{\underline{d} = 0.709}$$

$$D = \begin{bmatrix} - & 0{,}186 & 1 \\ 1 & - & 1 \\ 0{,}535 & 0{,}535 & - \end{bmatrix}$$

m= jumlah alternatif

Matrik dominan discordance

$$G = \begin{bmatrix} - & 0 & 1 \\ 1 & - & 1 \\ 0 & 0 & - \end{bmatrix}$$

6. Determining the Aggregate Dominance Matrix

Rumus : $e_{kl} = f_{kl} \times g_{kl}$

$$F = \begin{bmatrix} - & 1 & 0 \\ 0 & - & 0 \\ 1 & 1 & - \end{bmatrix} \quad \text{x} \quad G = \begin{bmatrix} - & 0 & 1 \\ 1 & - & 1 \\ 0 & 0 & - \end{bmatrix}$$

=

$$E = \begin{bmatrix} - & 0 & 0 \\ 0 & - & 0 \\ 0 & 0 & - \end{bmatrix}$$

$E_{12} = F_{12} \times G_{12} = 1 \times 0 = 0$

$E_{13} = F_{13} \times G_{13} = 0 \times 1 = 0$

$E_{21} = F_{21} \times G_{21} = 0 \times 1 = 0$

$E_{23} = F_{23} \times G_{23} = 0 \times 1 = 0$

$E_{31} = F_{31} \times G_{31} = 1 \times 0 = 0$

$E_{32} = F_{32} \times G_{32} = 1 \times 0 = 0$

7. Elimination of less favorable alternatives

- Matrix E gives the order of choice of each alternative, i.e. if $E_{kl} = 1$ then the alternative A_k is a better alternative than A_l
- If from matrix E there is no value of $E_{kl} = 1$, This means that there is no alternative chosen
- Next, ranking must be carried out using the value of **C_{kl} and D_{kl}**

7. Elimination of less favorable alternatives

Perform ranking using **value of C_{kl} and D_{kl}**

Alternatif	C_{kl}		D_{kl}	E	Rank
A_1	14	-	0,186	16,814	2
	4	-	1		
A_2	4	-	1	6	3
	4	-	1		
A_3	14	-	0,535	26,93	1
	14	-	0,535		

$$C = \begin{bmatrix} - & 14 & 4 \\ 4 & - & 4 \\ 14 & 14 & - \end{bmatrix}$$

$$D = \begin{bmatrix} - & 0{,}186 & 1 \\ 1 & - & 1 \\ 0{,}535 & 0{,}535 & - \end{bmatrix}$$

TERIMAKASIH.....